

[illegible]

3.3 V Buck-Boost

U6
TPS63031DSKR

SW

C4 10u 10V

C5 100n

L1 1.5uH

VIN VOUT 1

VINA

EN

PS/SYNC FB 10

L2

GND

TP1 3V3

C6 22u 10V

C7 22u 10V

TP2 GND

PS/SYNC high = forced PWM. FB wired to VOUT.
 Pad 11 = exposed thermal pad: bond to PGND/GND.
 Cout: retain >=20uF effective total at 3.3 V bias.
 L1 MPN: Murata DFE201612PD-1R5M=P2.

REFERENCE ONLY / PROFILE NOTES

MR—VERT: external magnetic pickup; optical LED OFF.
TR—VERT: onboard reflective optical tach; LED ON.
Both use internal XYZ MEMS and BLE to phone.

Report approximate RPM, 1/rev magnitude, phase, quality/stability and run-to-run trends.

REFERENCE / TROUBLESHOOTING ONLY.
Maintenance balancing remains with calibrated MicroVib / DynaVibe equipment.

Mechanical intent only; future PCB -50 x 40 mm.
No PCB generated. Keep antenna area clear.
Rigid mount near MEMS; baffled optical window on LEFT.
Validate phase delay and tach SNR on bench before use.

GPS MODULE (OPTIONAL) – IN-FLIGHT LOGGING

J4

+3V3

GPS_MODULE_3V3_UART

GPS_UART_RX

GPS_UART_TX

GPS_PPS

J4

1

2

3

4

5

J4: 1 VCC 3.3 V, 2 GND, 3 GPS TX → GPIO2 (UART1 RX), 4 GPS RX ← GPIO1 (UART1 TX), 5 PPS → GPIO13.
3.3 V logic module only (u-blox M8/M10 class, ~30 mA). Backup/sleep via USB over UART; no power switch.
UART1 keeps ROM boot output off the GPS. Ground speed only; not airspeed.

Three circuit diagrams for the USER INPUTS / STATUS section:

- ACQUIRE_BUTTON:** A circuit with a +3V3 supply connected to a 10k resistor (R8). The other end of R8 is connected to the ACQUIRE_BUTTON input. A switch (SW2) is connected between the ACQUIRE_BUTTON input and ground. A 10nF capacitor (C11) is connected between the ACQUIRE_BUTTON input and ground.
- BOOT:** A circuit with a +3V3 supply connected to a 10k resistor (R9). The other end of R9 is connected to the BOOT input. A switch (SW3) is connected between the BOOT input and ground. The switch is labeled "BOOT / SERVICE".
- STATUS_LED:** A circuit with a STATUS_LED supply connected to a 1k resistor (R10). The other end of R10 is connected to the STATUS_LED input. A green LED (LED1) is connected between the STATUS_LED input and ground. The LED is labeled "LED1 STATUS GREEN".
- SUPPLY_SENSE:** A circuit with a SYS_SW supply connected to a 100k resistor (R11). The other end of R11 is connected to the SUPPLY_SENSE input. A 100k resistor (R12) is connected between the SUPPLY_SENSE input and ground.

Active-low buttons; debounce in firmware.
 BOOT held low during reset enters download.
 Supply ADC: SYS_SW / 2; dead when OFF.
 One green status LED; patterns set by firmware.

ESP32-S3-MINI-1-N8

SW4 RESET

R4 10k

TP13 TP_EN

R5 10k

TP14 TP_BOOT

C10 1u 10V

C8 10u 10V

C9 100n

R6 22R

R7 22R

TP12 TP_USB_D-

TP11 TP_USB_D+

USB_CONN_D-

USB_CONN_D+

ESP32-S3-MINI-1-N8

EN 45

TXD0 43

RXD0 42

USB_D- 23

USB_D+ 24

IO21 25

IO26 26

IO33 28

IO34 29

IO35 31

IO36 32

IO37 33

IO38 34

IO39 35

IO40 36

IO41 37

IO42 38

IO43 39

IO44 40

IO45 41

IO46 42

IO47 43

IO48 44

GND 46

GPS_UART_TX 5

GPS_UART_RX 6

ACCEL_SDA 7

ACCEL_SCL 8

ACCEL_DRDY 9

MAG_TACH 10

OPT_TACH 11

OPT_LED_EN 12

ACQUIRE_BUTTON 13

STATUS_LED 14

SUPPLY_SENSE 15

GPS_PPS 16

IO0 40

IO1 41

IO2 42

IO3 43

IO4 44

IO5 45

IO6 46

IO7 47

IO8 48

IO9 49

IO10 50

IO11 51

IO12 52

IO13 53

IO14 54

IO15 55

IO16 56

IO17 57

IO18 58

IO19 59

IO20 60

IO21 61

IO22 62

IO23 63

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IO222 262

IO223 263

IO224 264

IO225 265

IO226 266

IO227 267

IO228 268

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IO230 270

IO231 271

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IO233 273

IO234 274

IO235 275

IO236 276

IO237 277

IO238 278

IO239 279

IO240 280

IO241 281

IO242 282

IO243 283

IO244 284

IO245 285

IO246 286

IO247 287

USB / BATTERY POWER SHARING

USB_VBUS

D2
PMEG4010CEH

Q3
AO3401A

BAT

R27
1k

C21
10u 10V

SW1
POWER

SYS

SYS_SW

USB supplies puck + charger; battery takes over without USB.
SW1 switches puck only. Q3: drain=BAT, source=SYS.

CS = VDDIO; SAO = GND -> 7-bit address 0x18.
 RES pad 7 MUST be GND. All three axes available.
 Firmware: high-performance, low-noise, 800/1600 Hz;
 INT1 data-ready timestamps; select bandwidth for RPM.
 Place near rigid mount; vertical axis set by orientation.

MAIN ROTOR MAG TACH

MAG- is common GND: isolated 2-wire VR pickup only.
 20k total limits input to <=3mA at +/-60V peak.
 Use two pulse-rated resistors; 1) internal input clamp.
 Mode 1: RC ~10us; timestamp leading FALLING edge.
 Polarity / phase offset set in profile. External twisted pair.

[illegible]

U3 = J_MAG: two plated wire holes only, 1 MAG+, 2 MAG-. No aircraft-specific board connector.

Sheet: /
File: BalancerREF.kicad_sch

Size: A2 Date: 2026-09-07

KiCad E.D.A. 10.0.3	Id: 1/1
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10	11
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